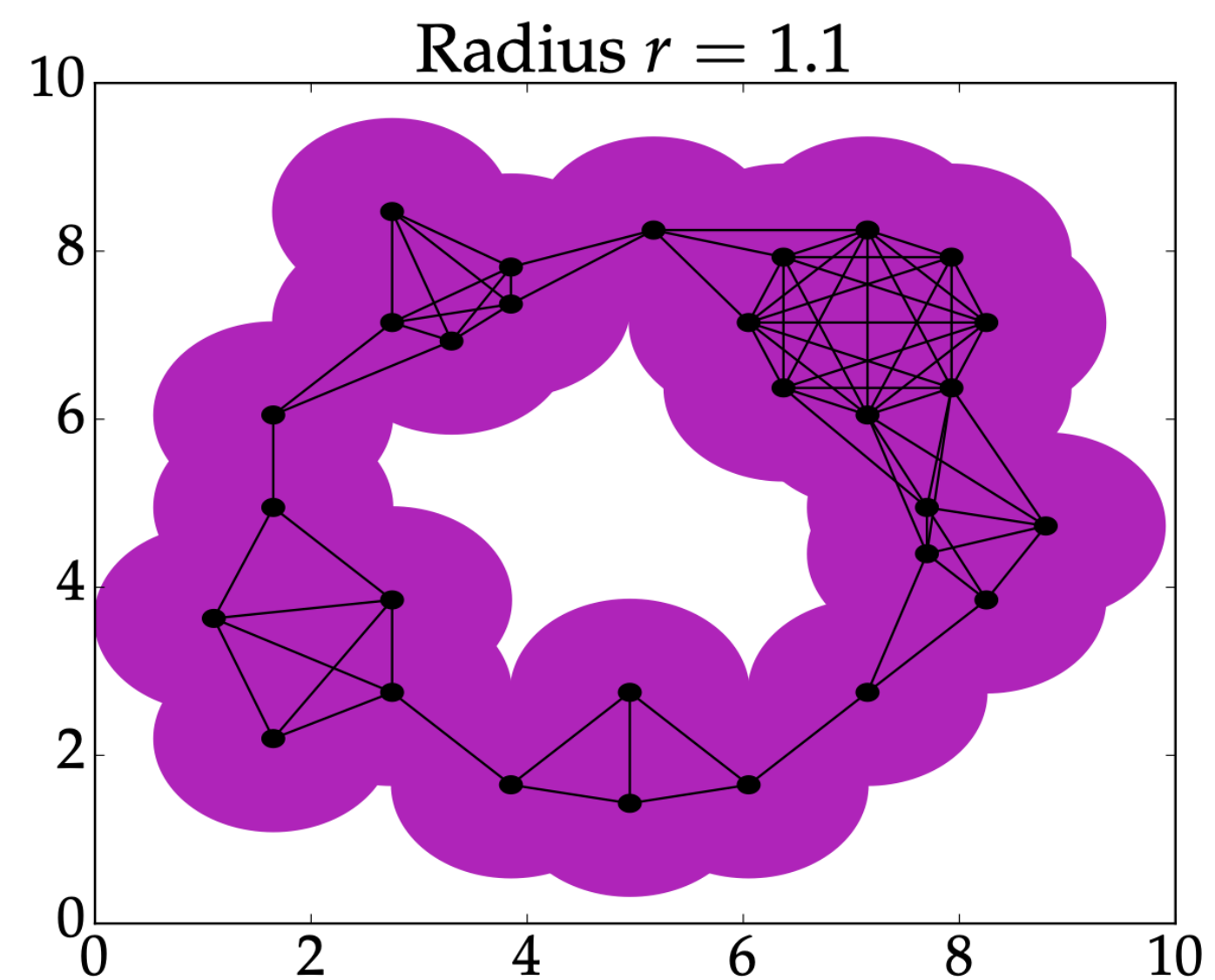
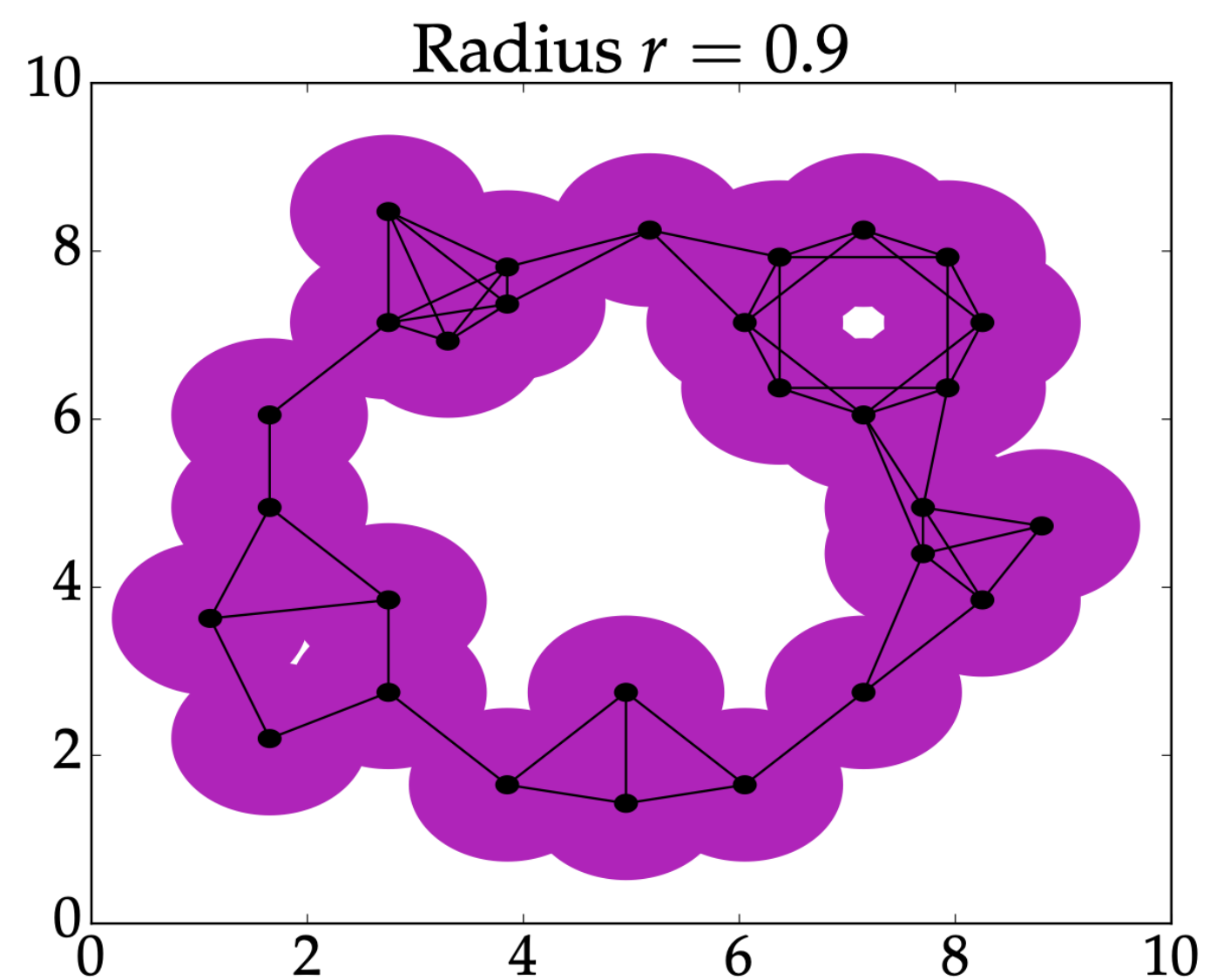
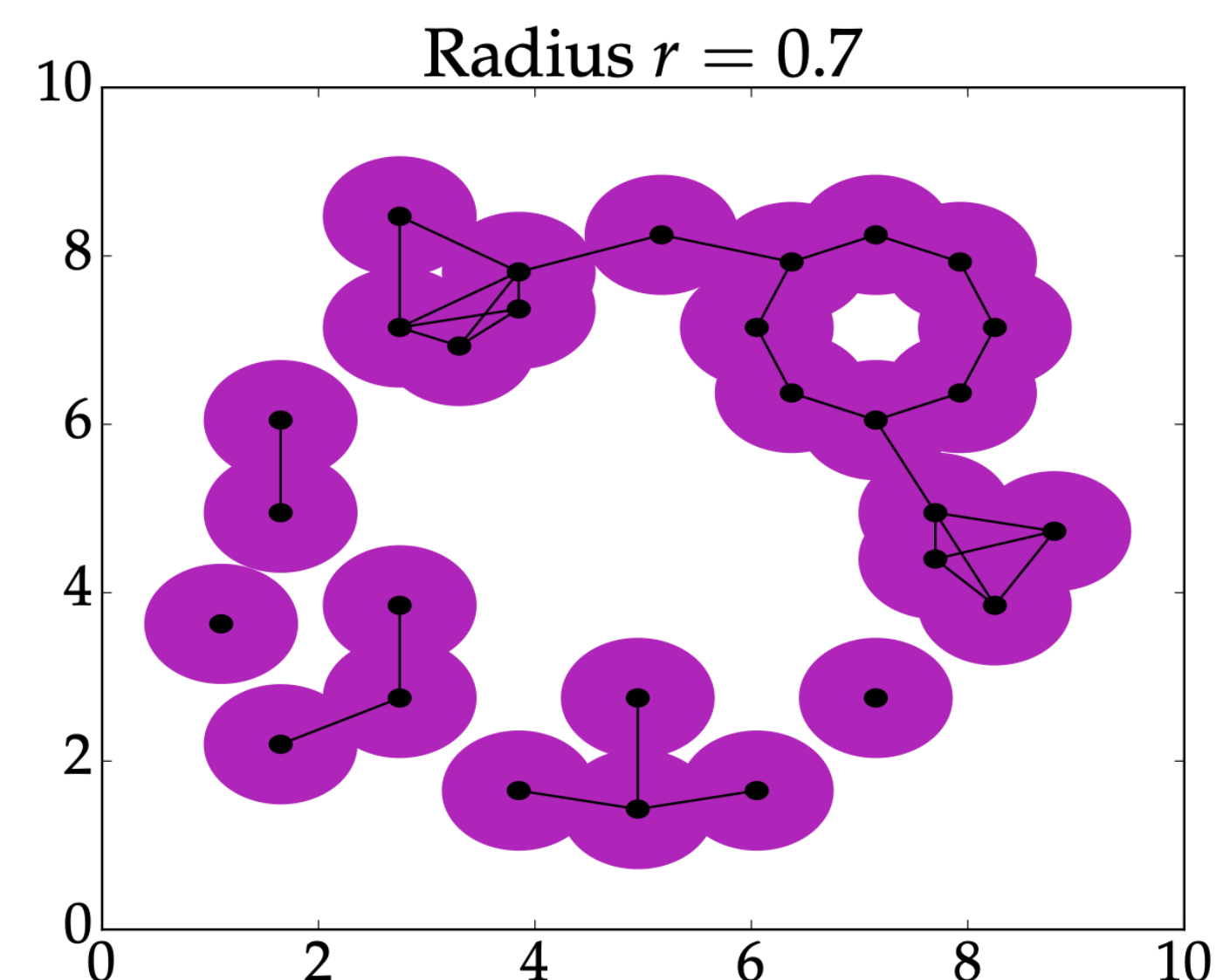
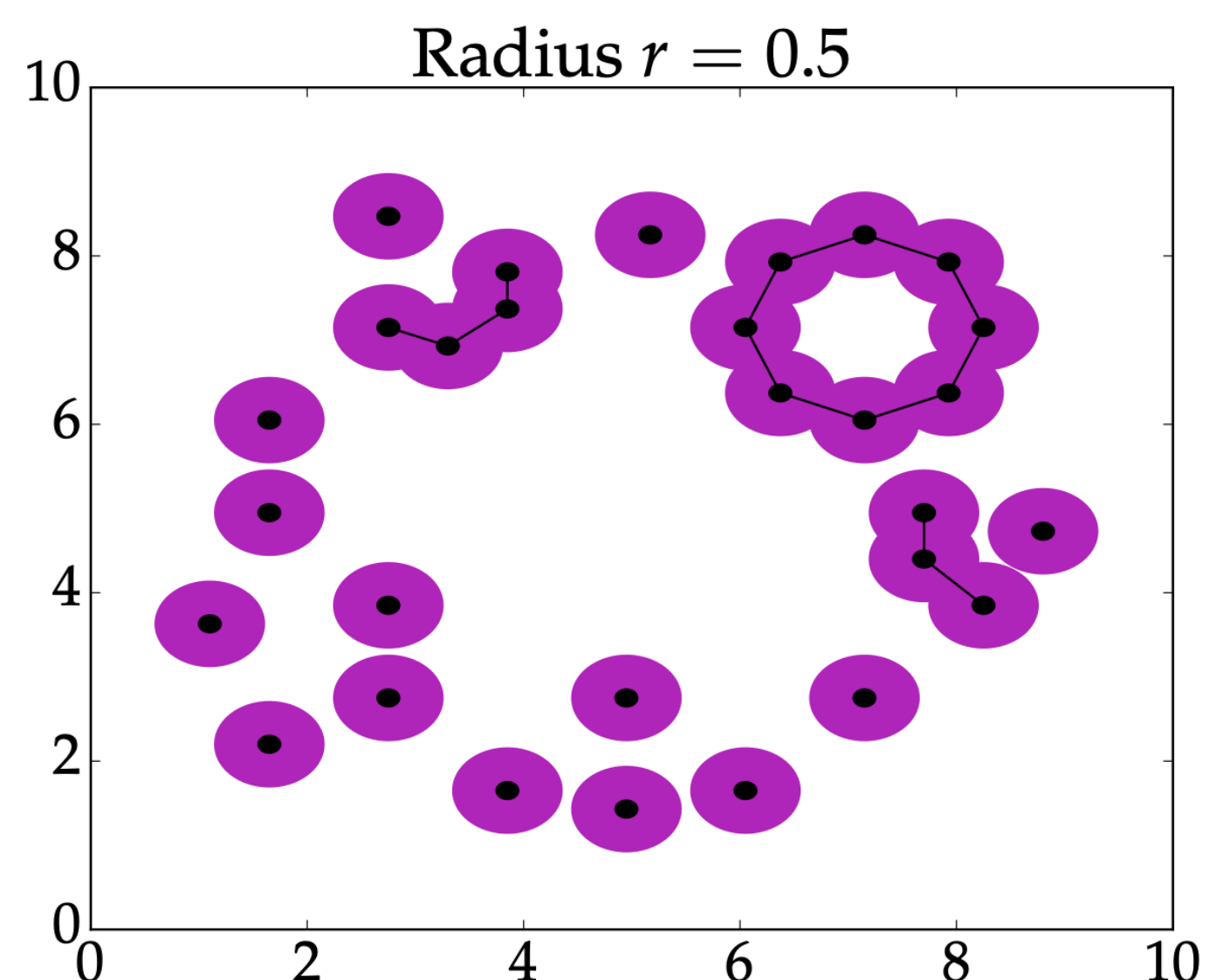
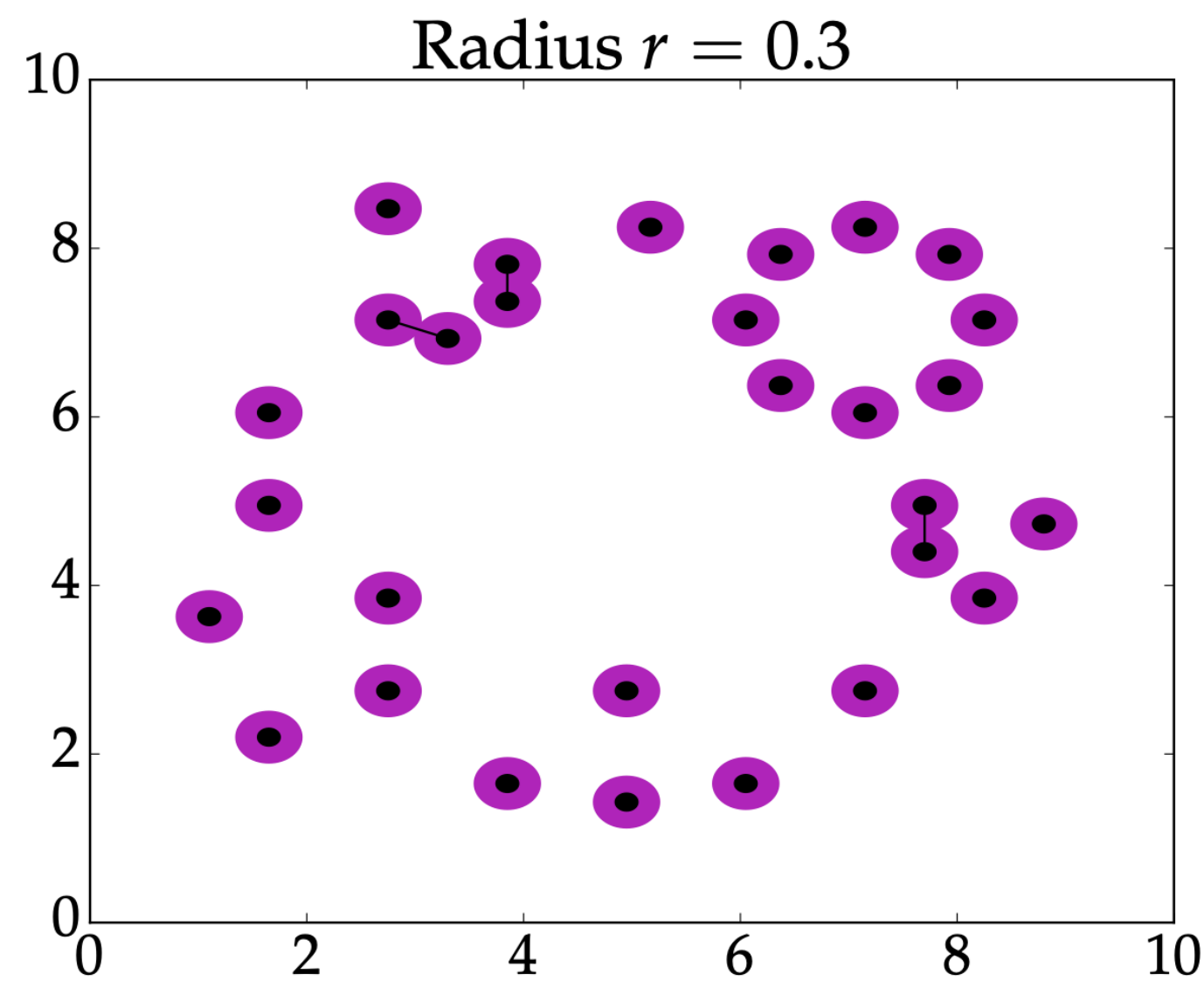


Here Comes the Homology

Will we get there?

Goal: what we want to study



Linear Algebra Review

Field

A field $(k, +, \cdot)$ is a set k with two operations $+$ and $\cdot$, such that for any $a, b, c \in k$:

- Closure $a+b \in k, a \cdot b \in k$.

- Commutativity

- $a+b = b+a$.

- $a \cdot b = b \cdot a$.

- Associativity

- $(a+b)+c = a+(b+c)$

- $(a \cdot b) \cdot c = a \cdot (b \cdot c)$

- Identity

- $0_k \Rightarrow a+0_k = a$.

- $1_k \Rightarrow a \cdot 1_k = a$.

$(\mathbb{Z}_p, +, \cdot)$ is a field
if p is prime.

- Inverses

- $\forall a, \exists -a, \text{ s.t. } a+(-a)=0_k$.

- $\forall a, \exists \frac{1}{a}, \text{ s.t. } a \cdot \frac{1}{a} = 1_k$.

- Distributivity

$$a \cdot (b+c) = a \cdot b + a \cdot c$$

Examples:

$(\mathbb{R}, +, \cdot)$ $x \in \mathbb{R}, \frac{1}{x} \in \mathbb{R} \text{ s.t. } x \cdot \frac{1}{x} = 1$.

$(\mathbb{Z}, +, \cdot)$ $2 \in \mathbb{Z}, \frac{1}{2} \notin \mathbb{Z}$.

$\mathbb{Z}_2 = \{0, 1\}$ $(\mathbb{Z}_2, +, \cdot)$

$+$	0	1
0	0 _k	1
1	1	0 _k

$\cdot$	0	1
0	0	0
1	0	1

Vector Space

A vector space over a field k is a set V with vector addition and scalar multiplication such that

- Associative $+$: $\uparrow^x$ $(v+w)+x = v+(w+x)$.
 $v, w \in V$, $x \in V$, $a \in k$.

- Commutative $+$: $v+w = w+v$.

- Identity:

$\downarrow \cdot +$ $0_v \in V$. $w + 0_v = w$.
 $w \in V$.

$\downarrow \cdot \cdot$ $1_k \in k$. $1_k \cdot v = v$.
 $v \in V$.

- Inverses $+$:
 $v \in V$, $\exists -v$ s.t. $v+(-v)=0_v$.

$$k \times V \rightarrow V$$

$\downarrow (a,v) \mapsto av$.

$$k \times k \rightarrow k$$

$\downarrow (a,b) \mapsto ab$.

- Scalar vs. field mult:

- Distributivity

$$(a+b) \cdot v = a \cdot v + b \cdot v.$$

$$a(v+w) = a \cdot v + a \cdot w.$$

Examples:

$$V = \mathbb{R}^2 \quad v = (1, 3).$$

$$k = \mathbb{R} \quad a=2, b=4.$$

$$av = 2 \cdot (1, 3) = (2, 6).$$

$$a \cdot b = 2 \cdot 4 = 8.$$

Basis

A basis for a vector space V is a collection of vectors $\{b_\alpha\}_{\alpha \in A}$ such that

- They are linearly independent and $\sum_{\alpha \in A} t_\alpha b_\alpha = 0 \Rightarrow t_\alpha = 0$ for all α .
 $t_\alpha \in k$ are scalar.
- They span V

Given any $v \in V$, $\exists \{t_\alpha\}$ s.t. $\sum_{\alpha \in A} t_\alpha b_\alpha = v$.

Ex $\mathbb{R}^2$, $b_1 = \{1, 0\}$, $b_2 = \{0, 1\}$.
 $\{b_1, b_2\}$.

if $a(1, 0) + b(0, 1) = (0, 0)$
 $a = b = 0$.

for any $(x, y) \in \mathbb{R}^2$

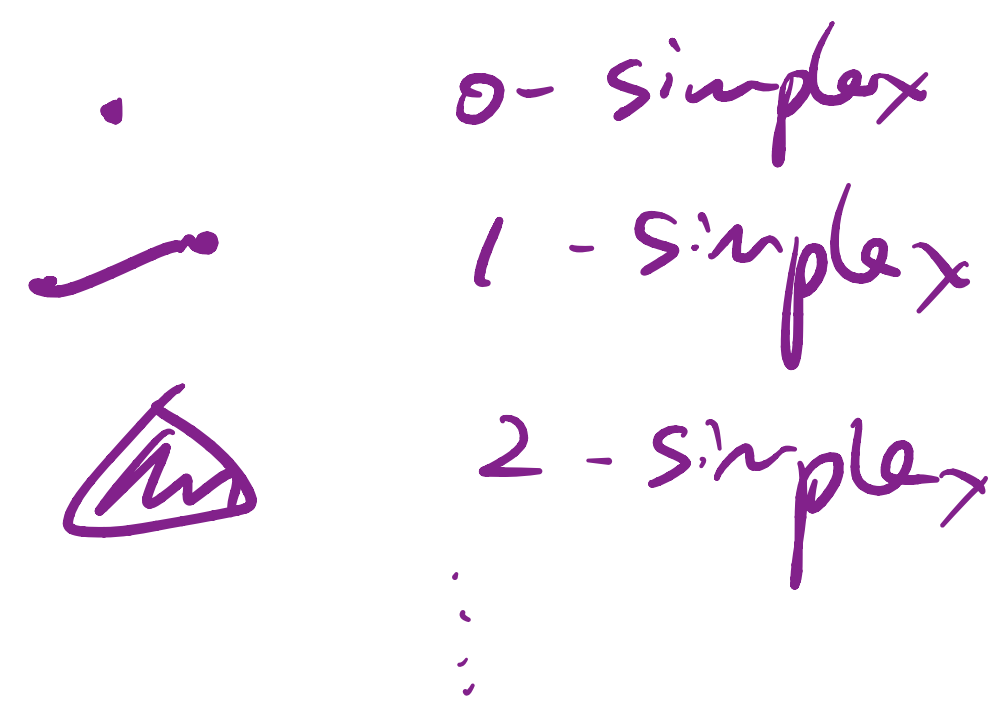
$$(x, y) = x \cdot b_1 + y \cdot b_2.$$

$\downarrow$
Generators: the set of vectors that spans the whole space.

$$\{(1, 0), (0, 1), (2, 3)\}$$

p -Chains

p -simplices.



Let K be a simplicial complex and fix a dimension p .

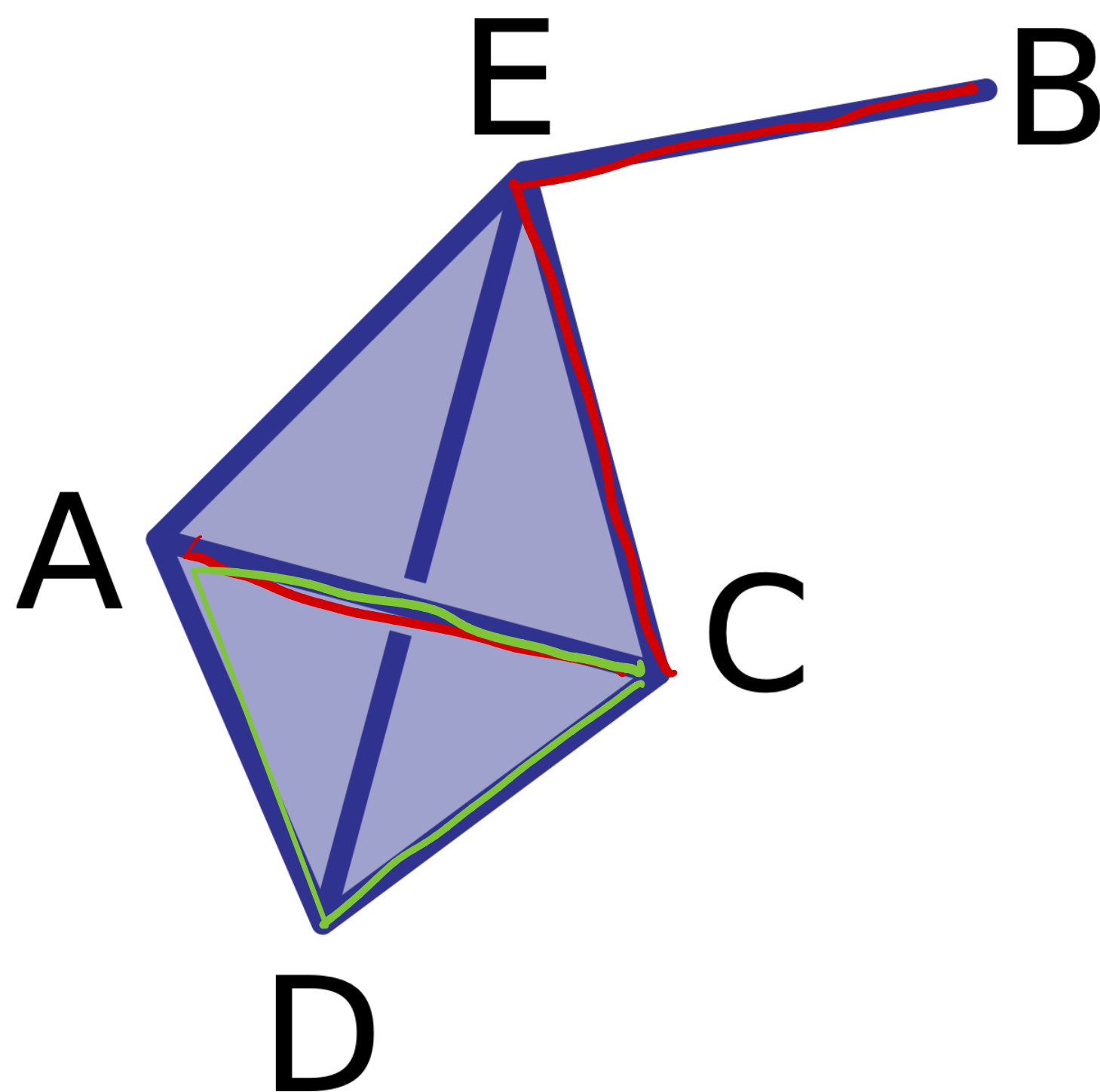
A p -chain is a finite formal sum of p -simplices in K , written $\alpha = \sum_i a_i \sigma_i \rightarrow p\text{-simplex}$.
 $1\text{-chain over } \mathbb{Z}_2 = \{0, 1\}$.

elements of $k = \mathbb{Z}_2$.

$$\alpha = AC + CE + BE$$

$$= 1 \cdot AC + 1 \cdot CE + 1 \cdot BE + 0 \cdot AE + 0 \cdot AD + 0 \cdot CD + 0 \cdot DE.$$

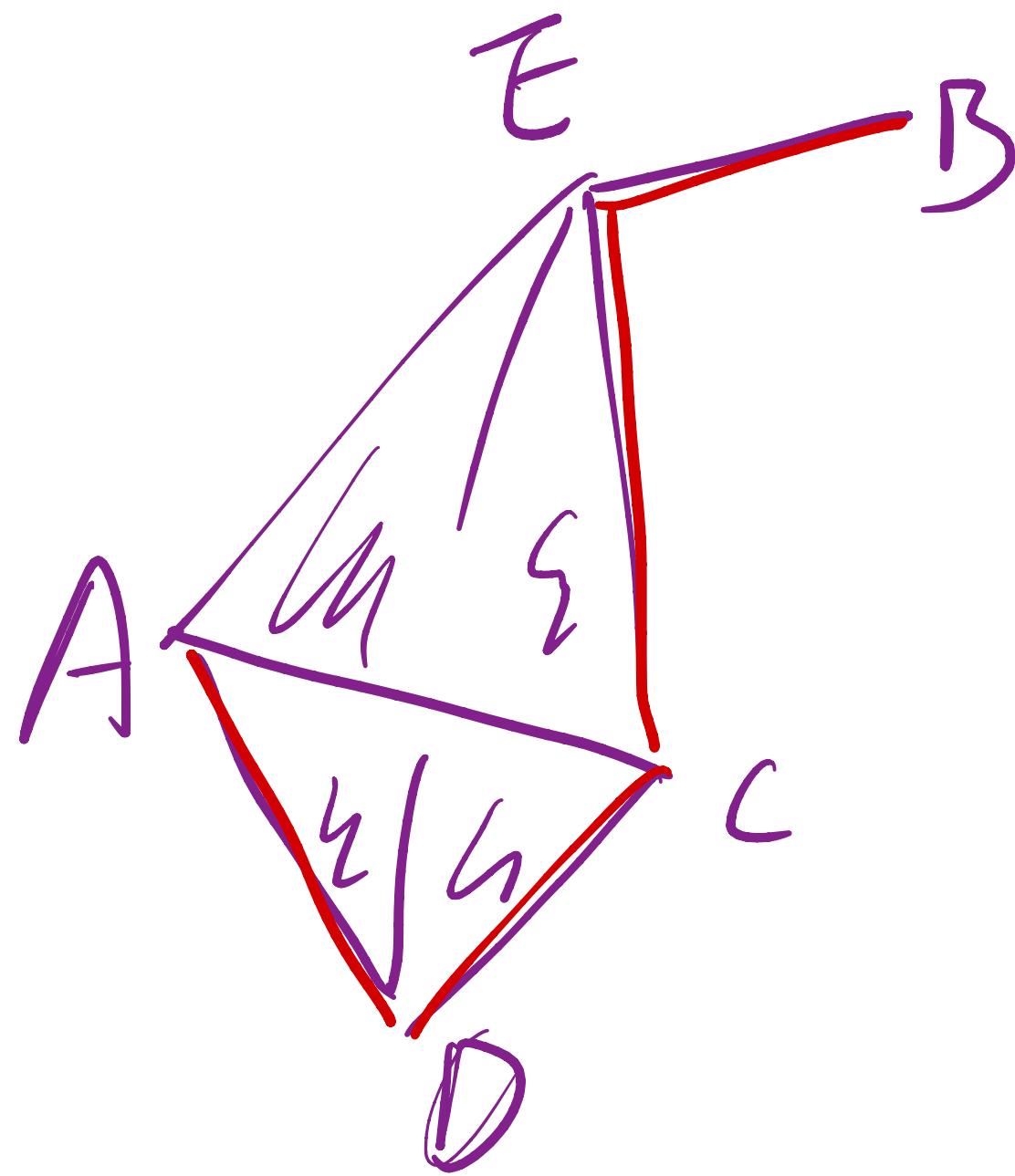
$$\beta = AD + AC + CD.$$



Addition of Chains

p -chains are added component-wise: if $\alpha = \sum a_i \sigma_i$ and $\beta = \sum b_i \sigma_i$, then

$$\alpha + \beta = \sum (a_i + b_i) \cdot \sigma_i$$



$$\alpha = AC + CE + BE.$$

$$\beta = AD + AC + CD.$$

$$\begin{aligned} \alpha + \beta &= 1 \cdot AC + 1 \cdot CE + 1 \cdot BE + 1 \cdot AD + 1 \cdot AC + 1 \cdot CD. \\ &= \underline{(1+1)} \cdot AC + CE + BE + AD + CD. \end{aligned}$$

p -chain.

Chain Group

The collection of p -chains with addition is called the p th-chain group (vector space), $C_p(K)$. $\leftarrow$ all linear combinations of p -simplices of K . $\downarrow$

Some checks: if $\alpha = \sum_i a_i \sigma_i$, $\beta = \sum_i b_i \sigma_i$, then $\alpha + \beta = \sum_i (a_i + b_i) \cdot \sigma_i \leftarrow$ basic vectors.

• Associative $(\alpha + \beta) + \gamma = \sum_i (a_i + b_i + c_i) \sigma_i = \alpha + (\beta + \gamma)$

• Commutative $\alpha + \beta = \beta + \alpha$. $\downarrow$ $a_i + b_i = b_i + a_i$

• Zero element $0 = \sum_i 0 \sigma_i$

• Inverses if $\alpha = \sum_i a_i \sigma_i$, then $-\alpha = \sum_i (-a_i) \sigma_i$, so $\alpha + (-\alpha) = 0$.

$$\mathbb{Z}_2 : -1 = 1 \quad 1+1=0$$

$$\alpha = AC + CE + BE.$$

$$\text{in } \mathbb{Z}_2 : -\alpha = \alpha. \quad \alpha + \alpha = 0.$$

$$\text{in } \mathbb{R} : -\alpha = -AC - CE - BE.$$

Linear Transformation

A linear transformation between two vector spaces V and W is a map $\underline{T} : V \rightarrow W$ such that the following hold:

- $T(v + w) = T(v) + T(w)$
- $T(a \cdot v) = a \cdot T(v)$

Suppose V has basis $\{b_1, \dots, b_n\}$.

for $v \in V$, $v = a_1 b_1 + \dots + a_n b_n$.

$$T(v) = T(a_1 b_1 + \dots + a_n b_n) \\ = a_1 T(b_1) + \dots + a_n T(b_n).$$

Matrix representation:

$$\begin{pmatrix} | & | & & | \\ T(b_1) & T(b_2) & \dots & T(b_n) \\ | & | & & | \end{pmatrix} \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = a_1 T(b_1) + \dots + a_n T(b_n).$$

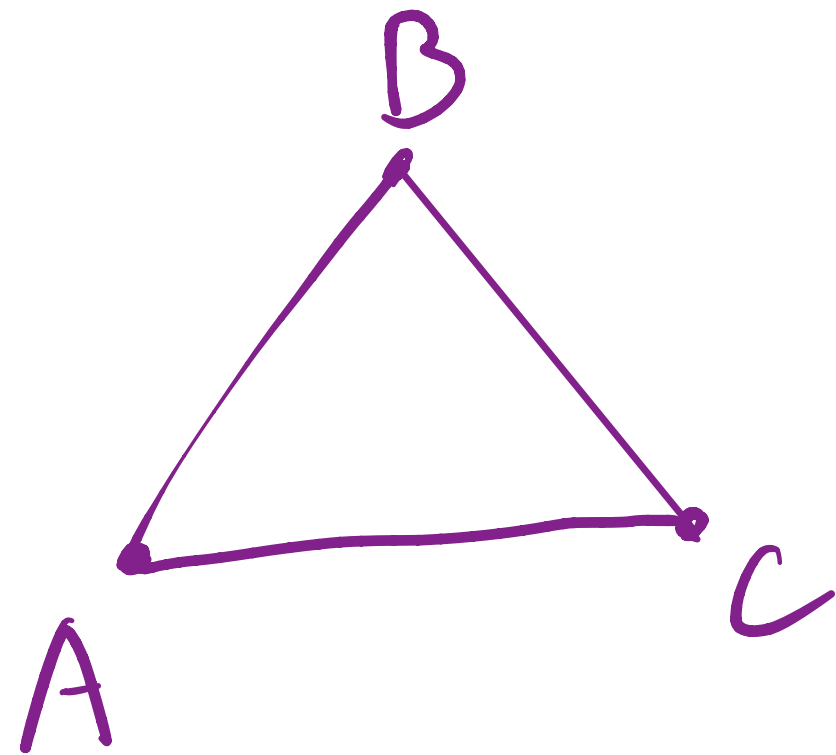
Boundary Maps

$$\partial_p : C_p(K) \longrightarrow C_{p-1}(K)$$

$$\sigma = [v_0, \dots, v_p] \longmapsto \sum_{j=0}^p [v_0, \dots, \widehat{v_j}, \dots, v_p]$$

element removal.

$$\sum_{j=0}^1 [v_0=A, v_1=B]$$



$$\text{Basis} : \{AB, AC, BC\}$$

$$C_1(K) \longrightarrow C_0(K)$$

$$\sigma_1 : AB \longmapsto A+B$$

$$\sigma_2 : AC \longmapsto A+C$$

$$\sigma_3 : BC \longmapsto B+C$$

$$\sigma_1 = [A, B]$$

$$= \sum_{j=0}^1 [\hat{v}_0=A, v_1=B] + [v_0=A, \hat{v}_1=B]$$

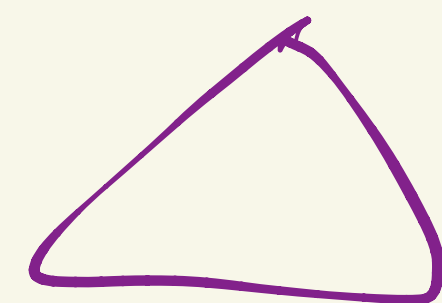
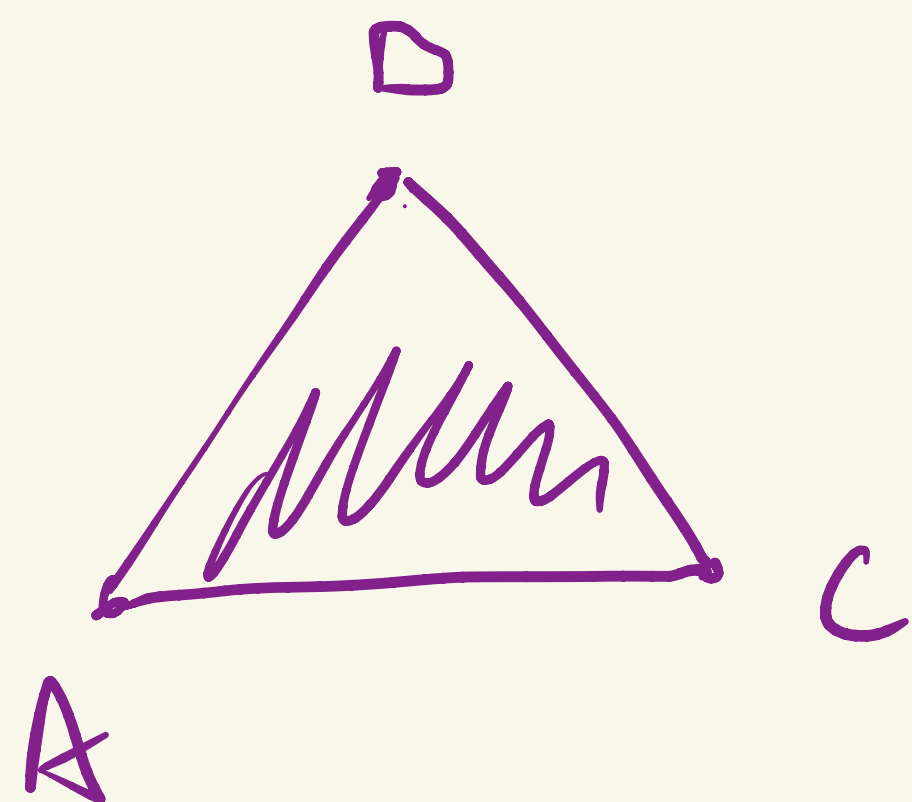
$$= [B] + [A]$$

$$= A+B$$

$$A+B + A+C + B+C$$

$$= 0$$

We assume $\mathbb{Z}_2$ coefficients from now on!



$$\begin{aligned}
 C_2(k) &\xrightarrow{\partial} C_1(k) \\
 ABC &\longrightarrow [\hat{A}, B, C] + [A, \hat{B}, C] \\
 &\quad + [A, B, \hat{C}] \\
 &= [B, C] + [A, C] + [A, B] \\
 &= \underline{BC + AC + AB}
 \end{aligned}$$

Checking Linearity

$$\partial_p(\alpha + \beta) = \partial_p(\alpha) + \partial_p(\beta); \quad \alpha = \sum a_i \sigma_i, \quad \beta = \sum b_i \sigma_i$$

$$\alpha = \overset{\downarrow}{A} \overset{\downarrow}{B} + \overset{\downarrow}{B} \overset{\downarrow}{C} \mapsto A+B+B+C = A+C = \partial_1(\alpha)$$

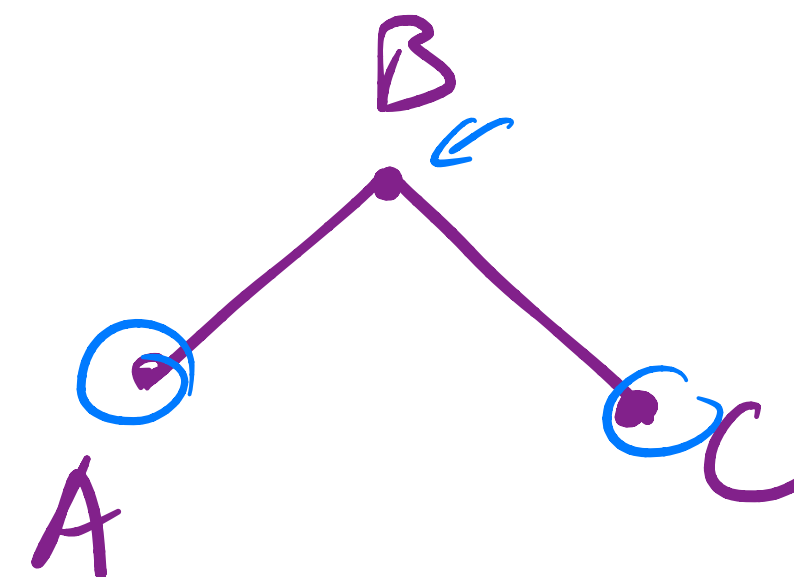
$$\beta = B C + A C \mapsto B+C+A+C = A+B = \partial_1(\beta)$$

$$\underline{\partial_1(\alpha) + \partial_1(\beta)} = A+C+A+B = C+B$$

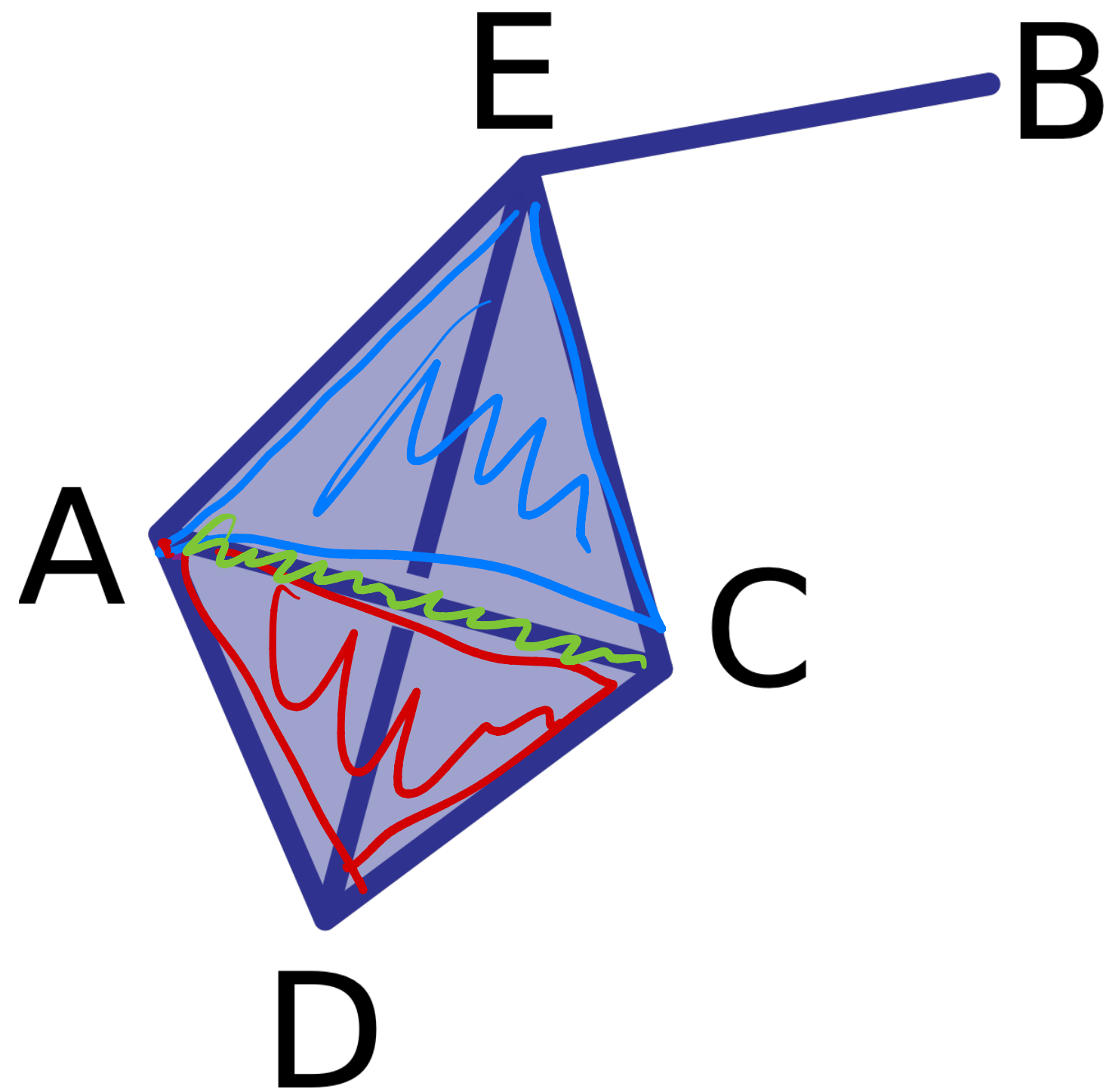
$$\alpha + \beta = AB + \underline{BC} + \underline{BC} + AC$$

$$= AB + AC$$

$$\underline{\partial_1(\alpha + \beta)} = A+B+A+C = B+C$$

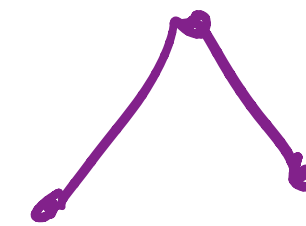


Example



- $\partial_1([a, e]) = A + E.$

- $\partial_1([a, e] + [b, e]) = A + \cancel{E} + B + \cancel{E} = A + B.$



- $\partial_1([a, e] + [c, e] + [c, d] + [a, d]) = A + E + C + E + C + D + A + D = 0.$

- $\partial_2([a, c, e] + [a, c, d]) =$

$$CE + AE + CD + AD.$$

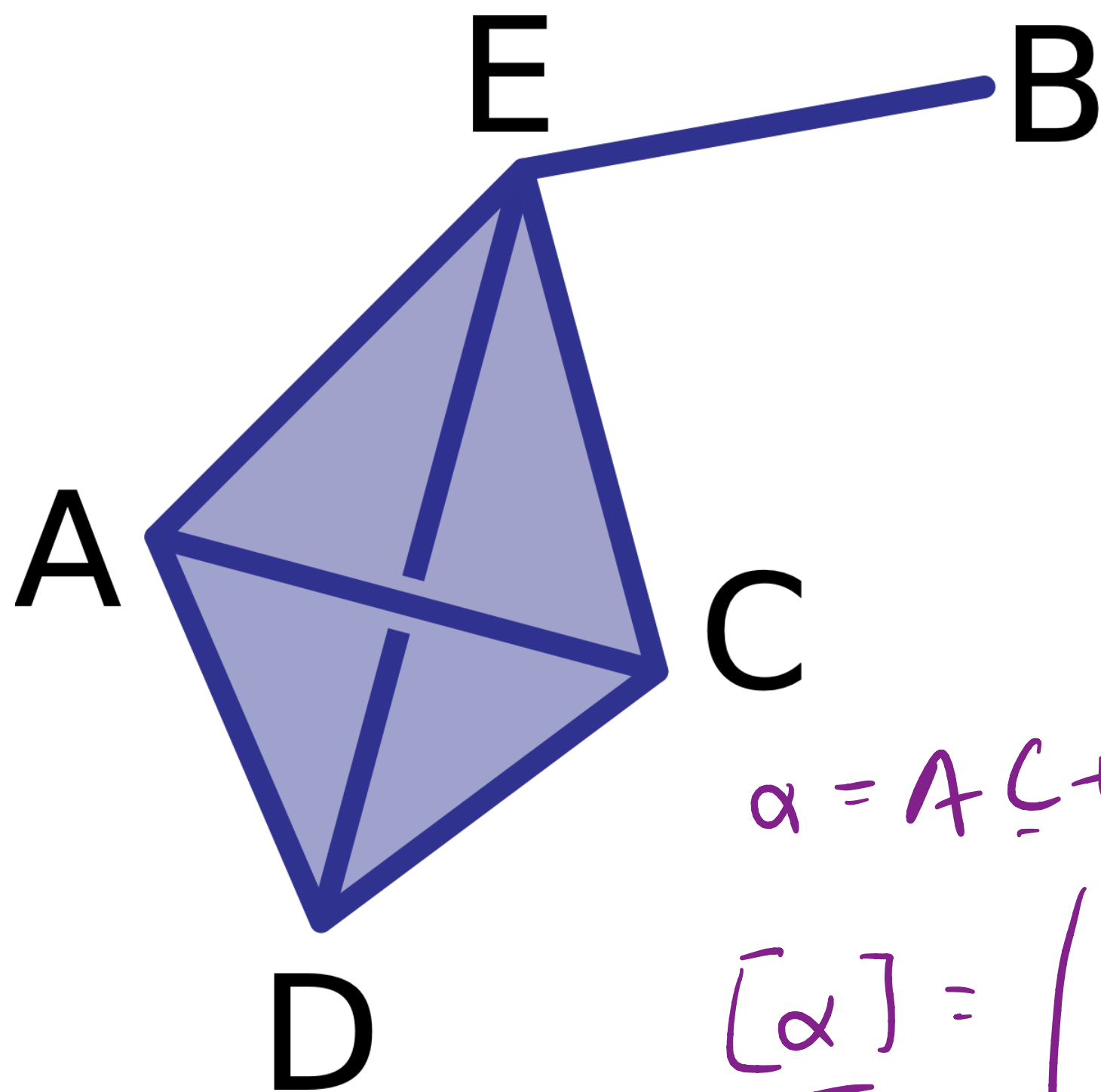
Matrix Representation

Basis of C_1 : $\{AC, AD, DC, BE, CE, DE, AE\}$

$\partial_1(K) : \underline{C}_1(K) \rightarrow \underline{C}_0(K)$ Basis of C_0 : $\{A, B, C, D, E\}$

$$AC \mapsto A + C$$

$$= 1 \cdot A + 0 \cdot B + 1 \cdot C + \dots$$



$$\alpha = AC + CE$$

$$[\alpha] = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \begin{matrix} AC \\ AD \\ AE \\ BE \\ CD \\ CE \\ DE \end{matrix}$$

	AC	AD	AE	BE	CD	CE	DE
A	1	1	1	0	0	0	0
B	0	0	0	1	0	0	0
C	1	0	0	0	1	1	0
D	0	1	0	0	1	0	1
E	0	0	1	1	0	1	1

$$\partial_1(K) \cdot [\alpha]$$

Cycles and Boundaries

Subspaces for a Linear Transformation

$$T: V \rightarrow W$$

- Image: $\text{Im}(T) = \{w \in W \mid \exists v \in V, T(v) = w\}.$
- Kernel $\text{Ker}(T) = \{v \in V \mid T(v) = 0\}.$

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$T(x, y) = x + y.$$

for any $r \in \mathbb{R}.$

$$T(r, 0) = r.$$

$$\text{Im}(T) = \mathbb{R}.$$

$$T(x, y) = x + y.$$

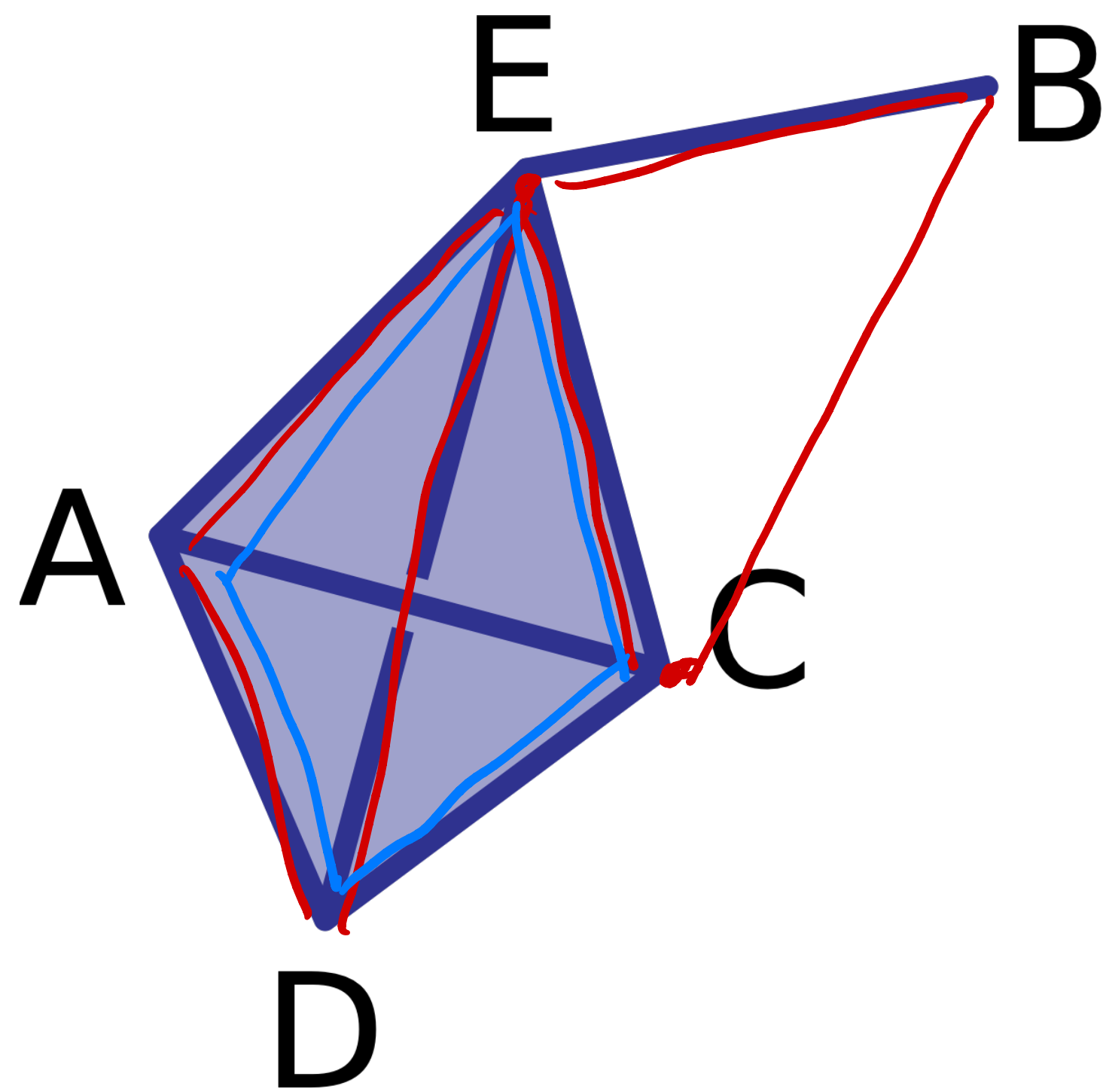
$$x + y = 0$$

$$x = -y.$$

Cycles

$$Z_p(K) = \ker(\partial_p). \quad C_{p+1}(K) \xrightarrow{\partial_{p+1}} C_p(K) \xrightarrow{\partial_p} C_{p-1}(K)$$

A chain in the kernel of ∂_p is called a p -cycle.



$$\alpha = AE + EC + CD + DA.$$

$$\partial_1(\alpha) = 0.$$

$$\alpha \in \ker(\partial_1) = Z_1(K).$$

$$\beta = AE + ED + AD + EC + BC + BE.$$

$$\partial_1(\beta) = 0.$$

$$\alpha + \beta = ED + CD + BC + BE.$$

$$\partial_1(\alpha + \beta) = 0.$$

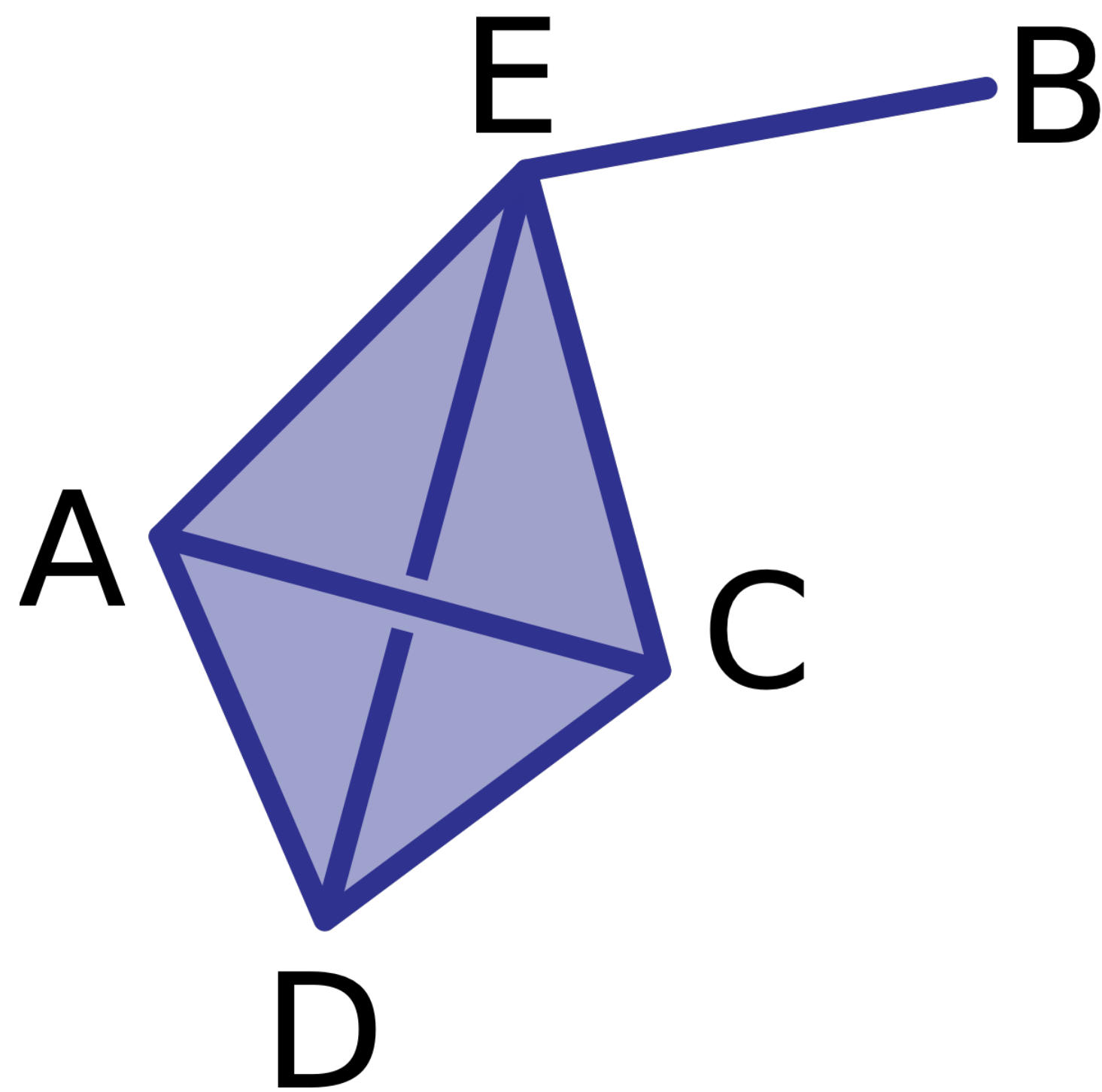
$$\alpha \in Z_1(K)$$

$$\beta \in Z_1(K)$$

$$\alpha + \beta \in Z_1(K).$$

The collection of p -cycles forms a subspace $\underline{Z_p(K)} \subseteq \underline{C_p(K)}$

What is a 2-Cycle?



$$Z_2(K) = \ker(\partial_2).$$

$$\alpha = ACE + ADE + DCE + ACD.$$

$$\partial_2: C_2(K) \rightarrow C_1(K).$$

$$ACE \mapsto \cancel{AC} + \cancel{CE} + \cancel{AE}$$

$$ADE \mapsto \cancel{AD} + \cancel{AE} + \cancel{DE}$$

$$DCE \mapsto \cancel{DC} + \cancel{CE} + \cancel{DE}$$

$$ACD \mapsto \cancel{AC} + \cancel{CD} + \cancel{AD}$$

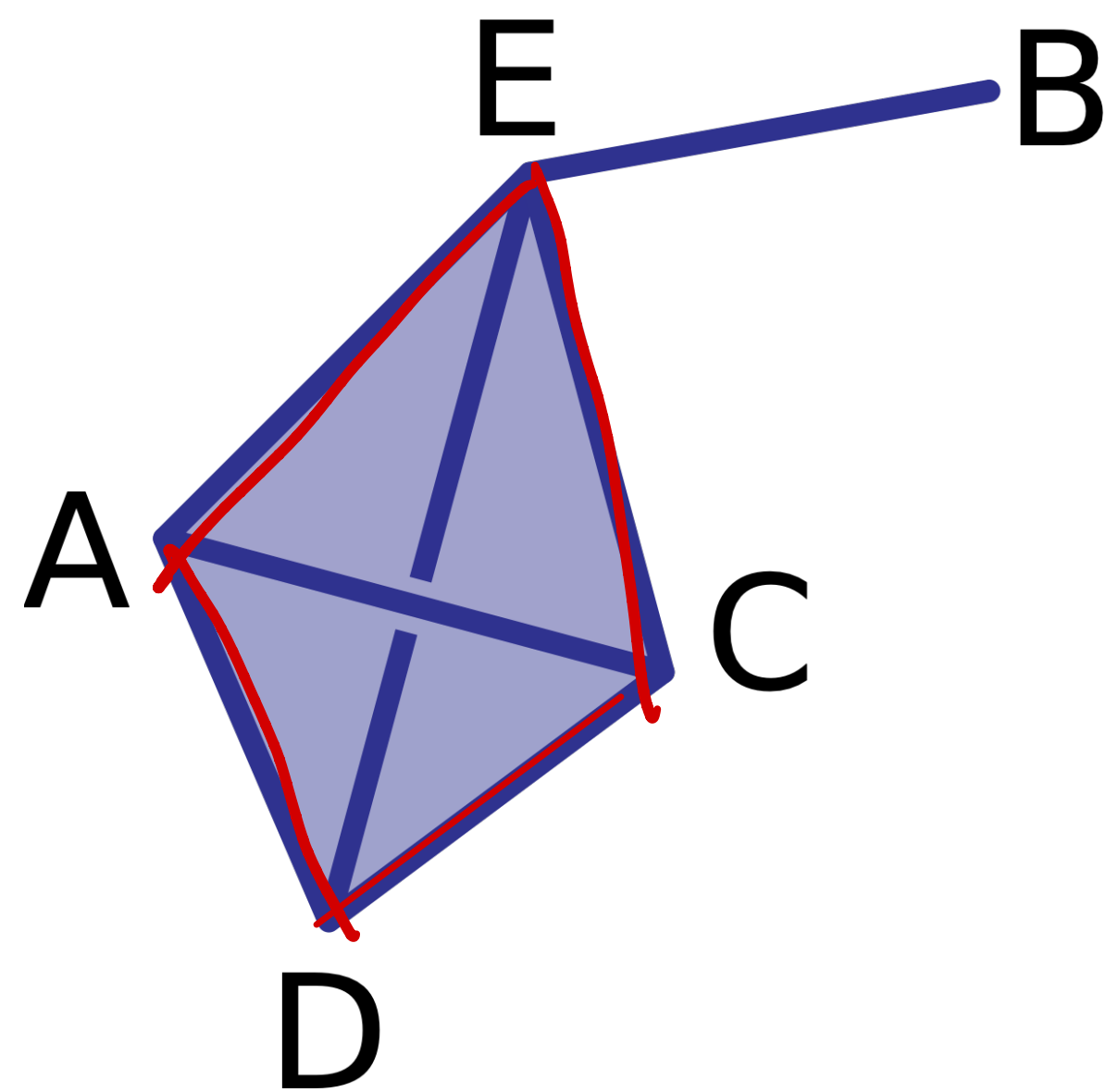
$$\partial_2(\alpha) = 0.$$

$$\alpha \in \ker(\partial_2) = Z_2(K).$$

Boundaries

$$C_{p+1}(K) \xrightarrow{\partial_{p+1}} \underline{C_p(K)} \xrightarrow{\partial_p} C_{p-1}(K)$$

A chain in the image of ∂_{p+1} is called a p -boundary



$$B_p(K) = \text{Im}(\partial_{p+1}).$$

$$B_p(K) = \{ \alpha \in C_p(K) \mid \exists \gamma \in C_{p+1}(K) \text{ with } \partial_{p+1}(\gamma) = \alpha \}$$

$$C_2(K) \xrightarrow{\partial} C_1(K) \xrightarrow{\partial} C_0(K).$$

$$ACD.$$

$$ACE.$$

$$\cancel{AC} + \cancel{CD} + AD.$$

$$\cancel{AC} + CE + \cancel{AE}.$$

$$\alpha \in B_1(K).$$

$$0.$$

$$0.$$

The collection of p -boundaries forms a subspace $B_p(K) \subseteq C_p(K)$

Nifty Trick

every boundary is a cycle.

Theorem: $\partial_p \partial_{p+1}(\alpha) = 0$ for every $(p+1)$ -chain α

$$C_{p+1}(K) \xrightarrow{\partial_{p+1}} C_p(K) \xrightarrow{\partial_p} C_{p-1}(K).$$

$$\alpha \mapsto \partial_{p+1}(\alpha) \mapsto \partial_p \partial_{p+1}(\alpha) = 0.$$

$\downarrow$
in $B_p(K)$,
in $Z_p(K)$.

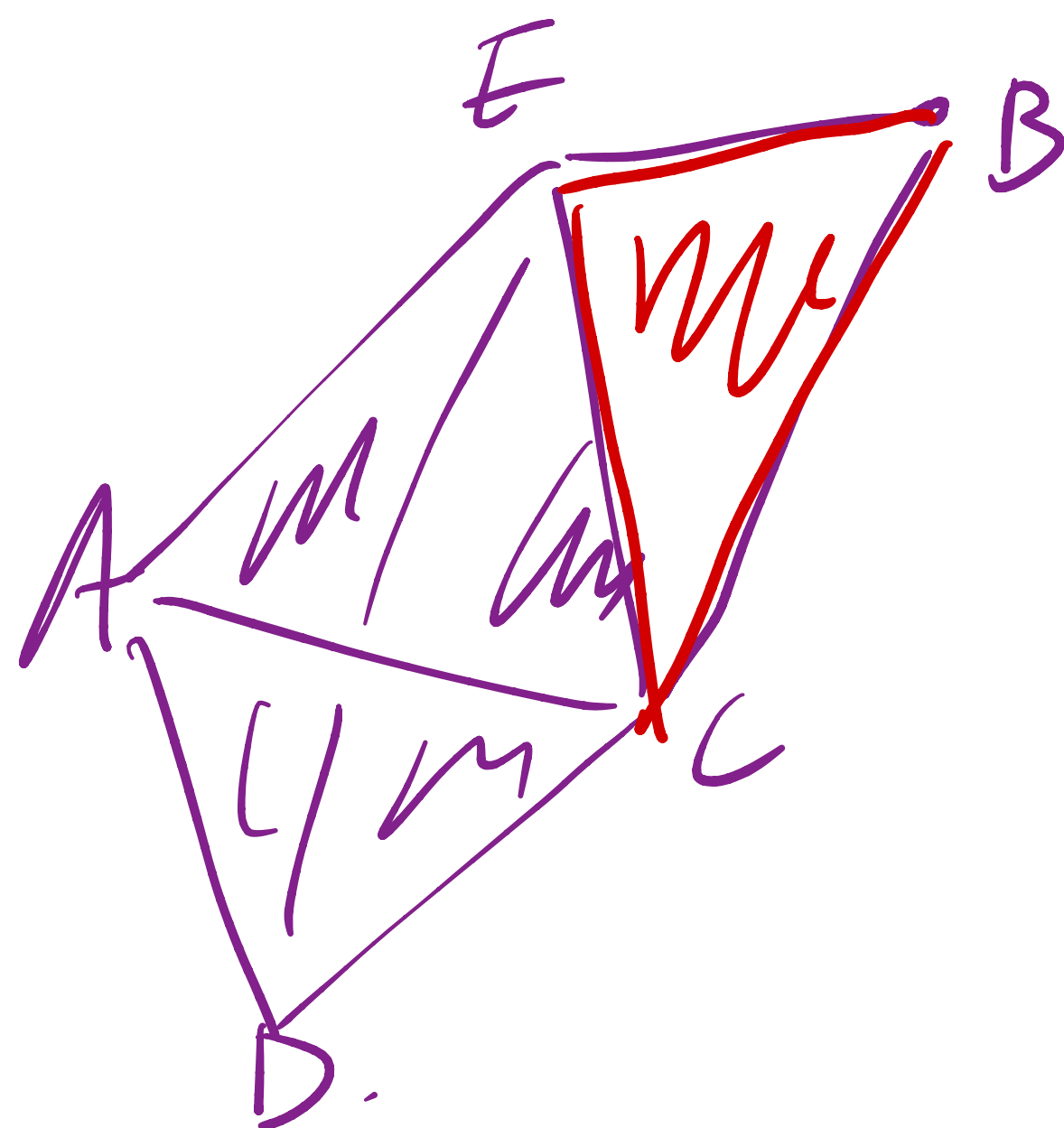
$$B_p(K) \subseteq Z_p(K) \subseteq C_p(K).$$

$$\gamma = EB + BC + CE.$$

$$\partial_1(\gamma) = 0.$$

$$\gamma \in Z_1(K).$$

$$\gamma \in B_1(K)?$$



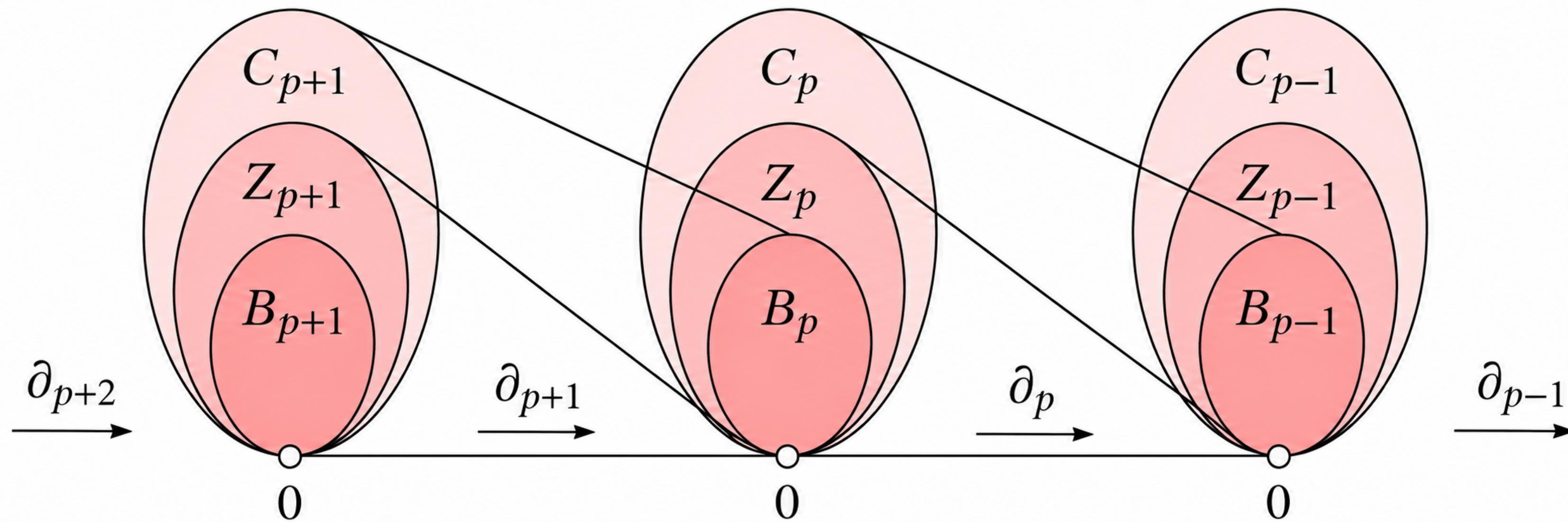
Translation

Every p -boundary is a p -cycle.

$$B_p(K) \subseteq Z_p(K) \subseteq C_p(K)$$

chain complex.
↓

$$C_{p+1}(K) \xrightarrow{\partial_{p+1}} C_p(K) \xrightarrow{\partial_p} C_{p-1}(K)$$



Chain Complex

$$\cdots \xrightarrow{\partial_{p+2}} C_{p+1}(X) \xrightarrow{\partial_{p+1}} C_p(X) \xrightarrow{\partial_p} C_{p-1}(X) \xrightarrow{\partial_{p-1}} \cdots \longrightarrow \underline{C_0(X)}.$$

$$\partial_p : C_p(K) \longrightarrow C_{p-1}(K)$$

$$\sigma = [v_0, \dots, v_p] \longmapsto \sum_{j=0}^p [v_0, \dots, \widehat{v_j}, \dots, v_p]$$

Homework

↙ Romano.

- Let K be the simplicial complex of a tetrahedron. Write a basis for the chain groups C_1 and C_2 ; the boundary groups B_1 and B_2 ; and the cycle groups Z_1 and Z_2 . Write the boundary matrix representing the boundary operator ∂_2 , with rows and columns representing bases of C_1 and C_2 , respectively.
- Consider a complex $K = \{A, B, C, D, E, AB, BC, AC, CD, AD, CE, DE, ABC, ACD\}$. Write a basis for the chain groups C_1 and C_2 ; the boundary groups B_1 and B_2 ; and the cycle groups Z_1 and Z_2 ; and write the matrix ∂_2 . Finally, identify a 1-cycle that is not a 1-boundary.

↖ Augustin.